

A

B

C

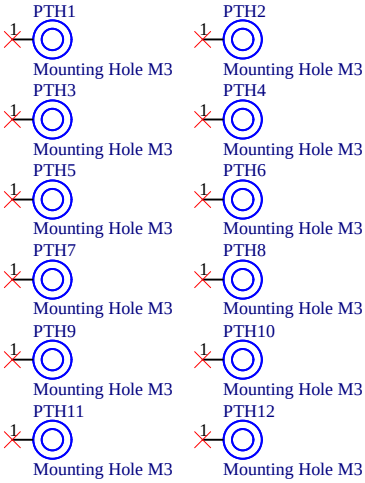
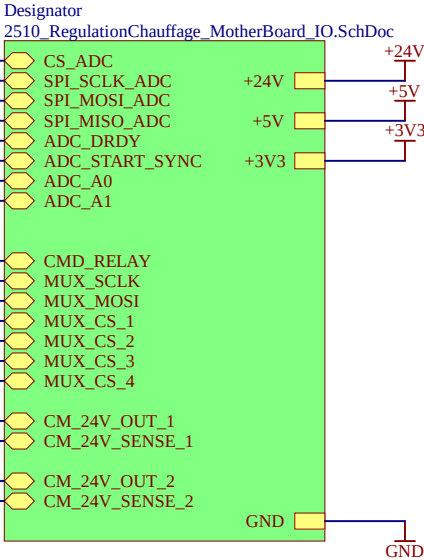
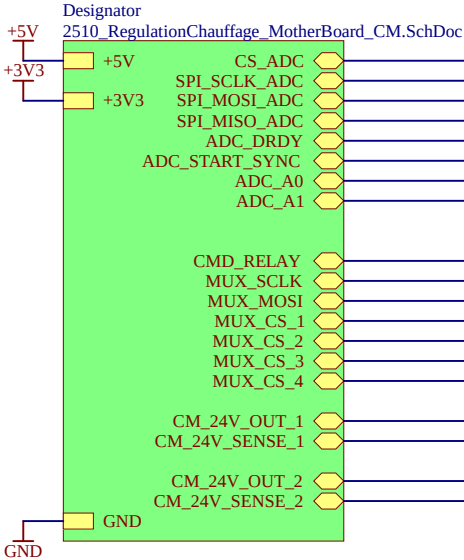
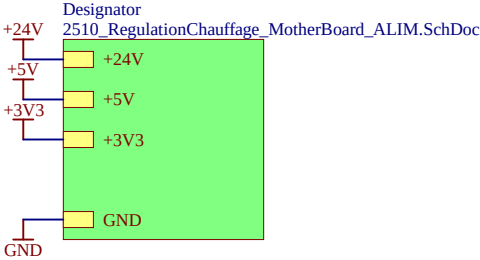
D

A

B

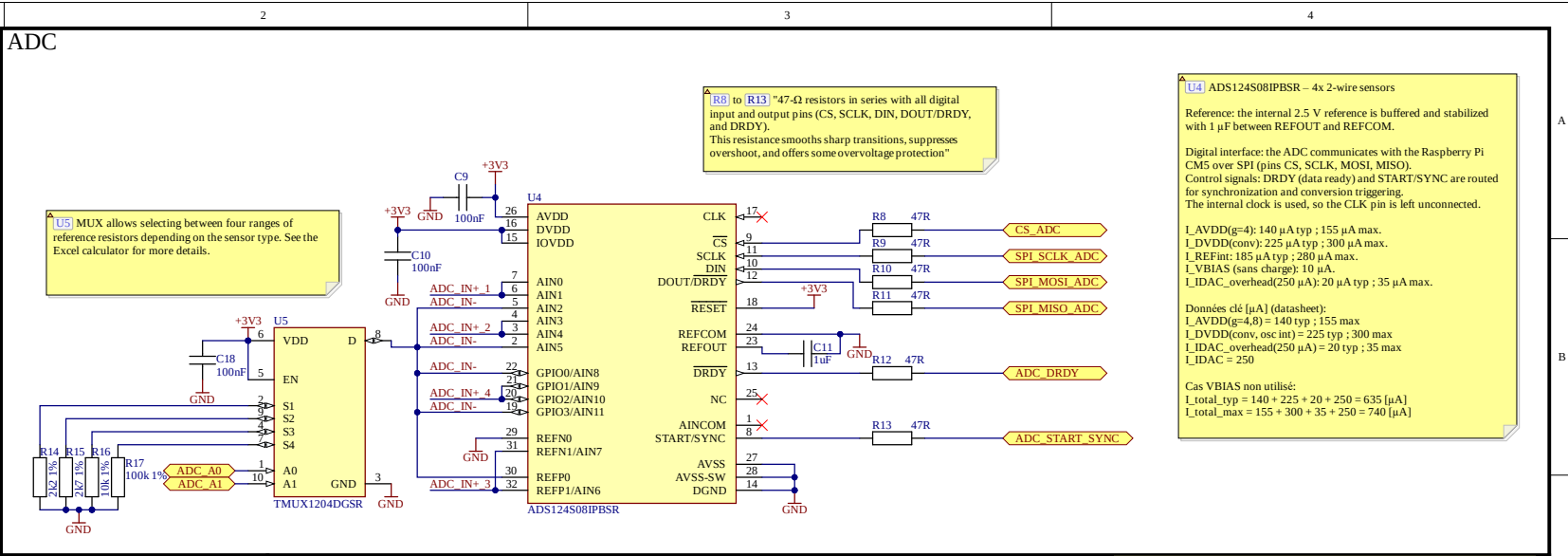
C

D



ObloSolution * * * * —	Mod.	a		Dessiné	LMS	19.08.2025	A4		Echelle : _____
		b		Contrôlé	*				
		c					Nb. feuilles 4	Feuille n° 1	
		d		Remplace					
Nom du projet		RegulationChauffage				 un temps d'avance	No 2510A		

I/O Sensor / Boiler



I/O DAUGHTER BOARD

J5

Pin	Signal
1	SENSOR IN+ 1
2	SENSOR IN- 1
3	ADC IN+ 1
4	ADC IN- 1
5	BOILER IN+ 1
6	BOILER IN- 1
7	CMD_RELAY
8	MUX_SCLK
9	MUX_CS_1
10	MUX_CS_2
11	MUX_MOSI

SBH11-PBPC-D10-ST-BK

J6

Pin	Signal
1	SENSOR IN+ 2
2	SENSOR IN- 2
3	ADC IN+ 2
4	ADC IN- 2
5	BOILER IN+ 2
6	BOILER IN- 2
7	CMD_RELAY
8	MUX_SCLK
9	MUX_CS_2
10	MUX_CS_1
11	MUX_MOSI

SBH11-PBPC-D10-ST-BK

J8

Pin	Signal
1	SENSOR IN+ 3
2	SENSOR IN- 3
3	ADC IN+ 3
4	ADC IN- 3
5	BOILER IN+ 3
6	BOILER IN- 3
7	CMD_RELAY
8	MUX_SCLK
9	MUX_CS_3
10	MUX_CS_2
11	MUX_MOSI

SBH11-PBPC-D10-ST-BK

J9

Pin	Signal
1	SENSOR IN+ 4
2	SENSOR IN- 4
3	ADC IN+ 4
4	ADC IN- 4
5	BOILER IN+ 4
6	BOILER IN- 4
7	CMD_RELAY
8	MUX_SCLK
9	MUX_CS_4
10	MUX_CS_3
11	MUX_MOSI

SBH11-PBPC-D10-ST-BK

24V - 3V3 I/O

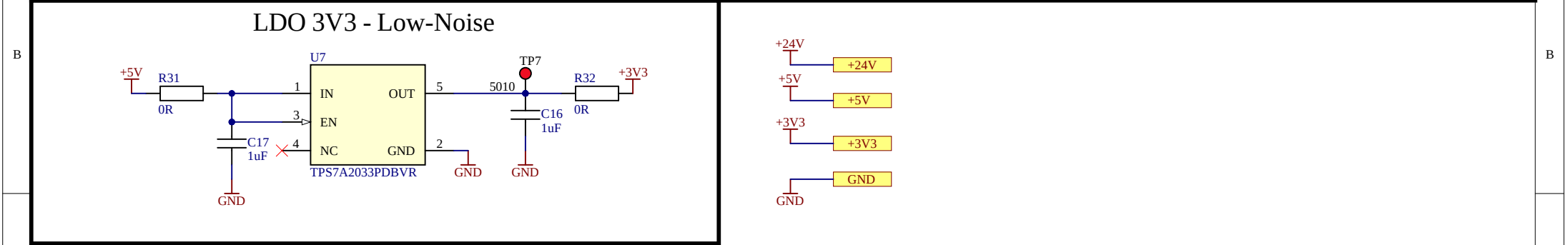
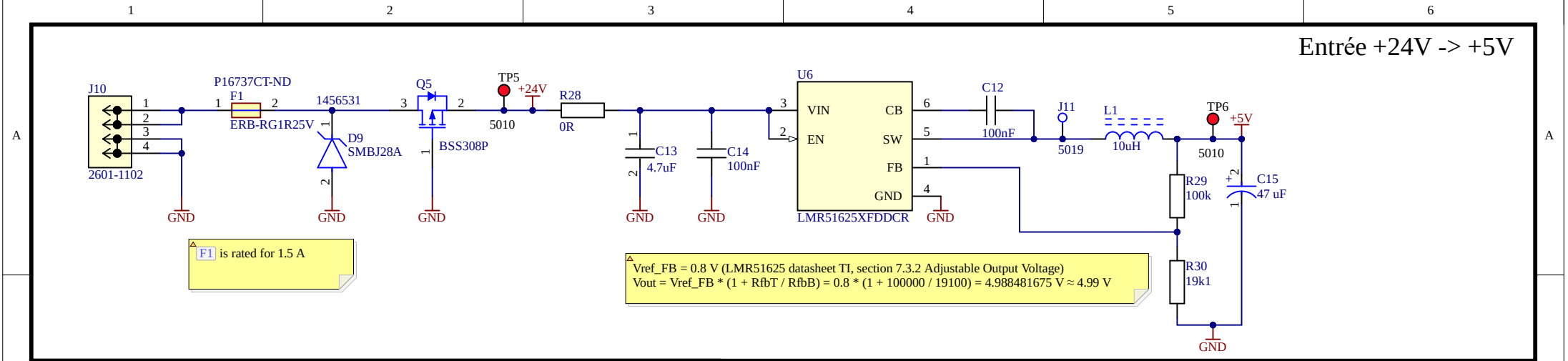
Single theoretical example, then multiplied. Two cases combined per channel:
 - Output worst case (pull-up to 24 V held low through 10 kΩ)
 - Input sense (24 V across divider 120 kΩ + 18 kΩ + 1 kΩ)

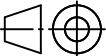
Output (per channel):
 $I_{out} = 24 / 10000 = 0.0024 \text{ A}$
 $P_{out} = I_{out} * V_{out} = 0.0024 * 0.0024 * 10000 = 0.0576 \text{ W}$

Input (per channel):
 $I_{in} = 24 / (120000 + 18000 + 1000) = 24 / 139000 = 0.000173 \text{ A}$
 $P_{in} = 24 * I_{in} = 24 * 0.000173 = 0.00415 \text{ W}$

Combined per channel (output + input):
 $I_{ch} = 0.0024 + 0.000173 = 0.002573 \text{ A}$
 $P_{ch} = 0.0576 + 0.00415 = 0.06175 \text{ W}$

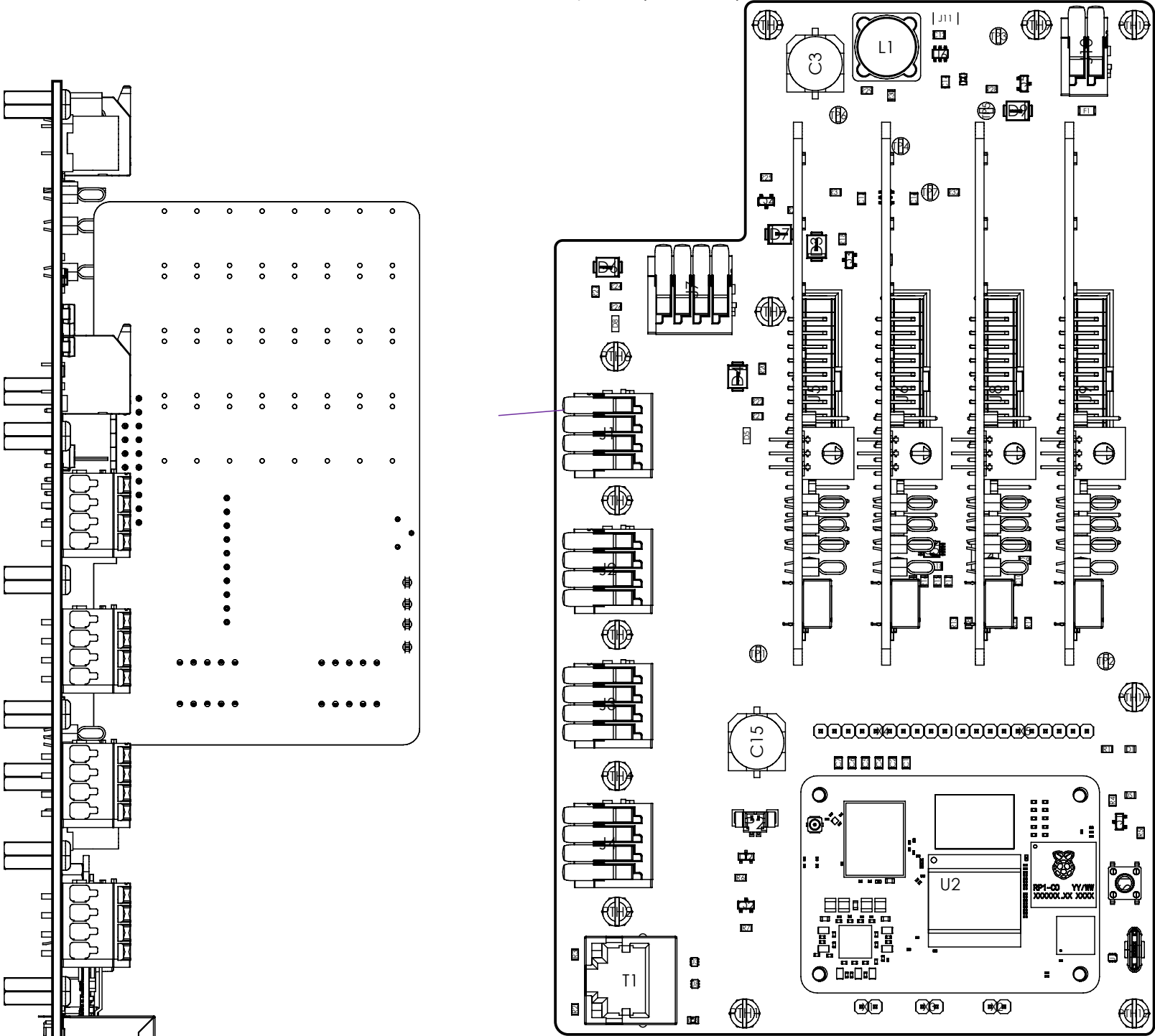
Totals for 2 identical channels:
 $I_{total} = 2 * 0.002573 = 0.005146 \text{ A}$
 $P_{total} = 2 * 0.06175 = 0.1235 \text{ W}$



ObloSolution * * * * —	Mod.	a		Dessiné	LMS	19.08.2025	A4		Echelle : _____
		b		Contrôlé	*				
		c					Nb. feuilles	Feuille n°	
		d		Remplace			4	4	
		Nom du projet RegulationChauffage					No 2510A		

Line #	Name	Description	Designator	Revision ID	Revision Status	Quantity	TargetPrice	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
	5019		J11			1	0	Keystone	5019	Volume Production	DigiKey	36-5019CT-ND	0.37	0.37
	CM5108032		U2			1	0	Amphenol Communications Solutions	10164227-1004A1RLF	Volume Production	DigiKey	609-10164227-1004A1RLFCT-ND	2.55	2.55
	IRLML2402		Q1, Q2			2	0	Infineon	IRLML2402TRPBF	Obsolete	Mouser	942-IRLML2402TRPBF	0.41	0.82
	Mounting Hole M3		PTH1, PTH2, PTH3, PTH4, PTH5, PTH6, PTH7, PTH8, PTH9, PTH10, PTH11, PTH12			12	0							
	second connector CM5		U2-21			1	0	Amphenol Communications Solutions	10164227-1004A1RLF	Volume Production	DigiKey	609-10164227-1004A1RLFCT-ND	2.55	2.55
	CONN JACK 1PORT PCB	1 Port RJ45Trou traversant 10/100 Base-T, AutoMDIX	T1			1	0				DigiKey	5923-SI-46001-F-ND		
	TPS7A2033PDBVR	300mA ultra-low-noise low-IQ low-dropout (LDO) linear regulator with high PSRR 5-SOT-23-40 to 125	U7			1	0	Texas Instruments	TPS7A2033PDBVR	Volume Production	Mouser	595-TPS7A2033PDBVR	0.31	0.31
	LMR51625XFDCCR	Abaisseur de tension Régulateur à découpage IC Positif Réglable 0,8V 1 Sortie 2,5A SOT-23-6 mince, TSOT-23-6	U6			1	0	Texas Instruments	LMR51625XFDCCR	Volume Production	DigiKey	296-LMR51625XFDCCRCT-ND	2.03	2.03
	BSS308P	Canal P 30 V 2 A (Ta) 500mW (Ta) Montage en surface PG-SOT23	Q5			1	0	Infineon	BSS308PEH6327XTSA1	Volume Production	Farnell	2432719	0.48647	2.43
	47 uF	CAP ALUM 47UF 20% 50V SMD	C3, C15			2	0	Panasonic	EEFT1H470AP	Volume Production	Farnell	1868425	0.51766	1.04
	0.1 uF	CAP CER 0.1 uF 10V X5R 0805	C8			1	0	Yageo Group	CC0805KRX7R9B8104	Volume Production	Farnell	3019949	0.03006	0.30062
	4.7uF	CAP CER 4.7UF 10V X5R 0805	C13			1	0	Yageo Group	CC0805KKX5R6B8475	Volume Production	DigiKey	311-1863-1-ND	0.2	0.2
	1uF	CAPC 1uF X5R 25V 10% 2012 (0805 INCH)	C11, C16, C17			3	0	TDK	C2012X5R1E105K125AA	Volume Production	DigiKey	445-1419-1-ND	0.18712	0.56135
	100nF	CAPC 100nF X7R 50V 10% 2012 (0805 INCH)	C1, C2, C4, C5, C6, C7, C9, C10, C12, C14, C18			11	0	KYOCERA AVX	08055C104KAT2A	Volume Production	Farnell	499687	0.03842	0.42261
	SBH11-PBPC-D10-ST-BK	Connector Header Through Hole 20 position 0.100 (2.54mm)	J5, J6, J8, J9			4	0	Sullins	SBH11-PBPC-D10-ST-BK	Volume Production	DigiKey	S9172-ND	0.41	1.64
	2601-1102	Description				1	0	WAGO	2601-1102		Farnell	4126046	1.91	1.91
	2601-1104	Description	J1, J2, J3, J4, J7			5	0	WAGO	2601-1104		Farnell	4126048	3.54	17.71
	ADS124508IPBSR	IC ADC 24BIT SIGMA-DELTA 32LQFP	U4			1	0	Texas Instruments	ADS124508IPBSR	Volume Production	DigiKey	296-45170-1-ND	11.78	11.78
	2.54	JUMPER 2.54 1x02	X1, X2, X3			3	0	Würth Elektronik	61303611121		Mouser	710-61303611121	1.38	4.14
	LTST-C170KRKT	LED Red, 75 mW, 54 mcd, -55 to 85 degC, 2-Pin SMD, RoHS, Tape and Reel	DS1			1	0	Lite-On	LTST-C170KRKT	Volume Production	DigiKey	160-1415-1-ND	0.16	0.16
	53398-0271	Male Header, Pitch 1.25 mm, 1 x 2 Position, Height 4.7 mm, -40 to 85 degC, RoHS, Tape and Reel	P2			1	0	Molex	53398-0271	Volume Production	DigiKey	WM7606CT-ND	0.54	0.54
	ERB-RG1R25V	Micro Chip Fuse, Size [L—W] (EIA) (mm): 3.2Å—1.6 (EIA:1206), Rated Current (A): 1.25, Internal R [at 25Å,f max.] (mÅ, J): 175, Rated Voltage (Open Circuit Voltage) [DC] (V): 63, Interrupting Rating (A): 50, Category Temp. Range (Å°C): -40 to +125	F1			1	0				Digikey	P16737CT-ND		
	IRLML6344	MOSFET N-CH 30V 5A MICRO3/SOT23	Q3, Q4			2	0	Infineon	IRLML6344TRPBF	Volume Production	Farnell	1857299	0.39417	1.97
	TMUX1204DGSR	MULTIPLEXER	U5			1	0	Texas Instruments	TMUX1204DGSR	Volume Production	Mouser	595-TMUX1204DGSR	0.57	0.57
	Raspberry Pi SC1570	Ordinateur monocarte 2.4GHz 4 Cœur 8Go/0Go RAM ARM® Cortex®-A76	U2[Externe]1			1	0	Raspberry Pi	CMX008016		Newark	20AM3761	80	80
	2.54	PIN HEADER 2.54 1x10	X4, X5			2	0	Würth Elektronik	61300411121		Mouser	710-61300411121	0.17	0.34
	TPD4E1U06DCKR	Quad 0.8-pF, 5.5-V, Å±15-kV ESD protection diode in SOT-23 & SC70 packages 6-SC70 -40 to 125	U1, U3, U8			3	0	Texas Instruments	TPD4E1U06DCKR	Volume Production	DigiKey	296-39347-1-ND	0.58	1.74
	SC1752	RASPBERRY PI CM5 PASSIVE COOLER	U2[Externe]1			1	0	Raspberry Pi	SC1752		DigiKey	2648-SC1752-ND	5	5
	0R	RES 0ohm ±1% 0.125W 2012 (0805 INCH)	R28, R31, R32			3	0	Vishay	CRCW0805000020EA	Volume Production	Newark	52K9721	0.022	0.066
	1k0	RES 1.0kohm ±1% 0.125W 2012 (0805 INCH)	R21, R26			2	0	Yageo Group	RC0805FR-071KL	Volume Production	Newark	82K7536	0.004	0.008
	1k5	RES 1.5kohm ±1% 0.125W 2012 (0805 INCH)	R2, R3			2	0	Yageo Group	RC0805FR-071K5L	Volume Production	DigiKey	311-1.50KCRCT-ND	0.01169	0.11695
	2k2 1%	RES 2.2K OHM 1% 1/8W 0805 RC0805FR-072K2L	R14			1	0	Yageo Group	RC0805FR-072K2L	Volume Production	Farnell	9237534	0.01222	0.12224
	2k7 1%	RES 2.7K OHM 1% 1/8W 0805 RC0805FR-102K7L	R15			1	0	Yageo Group	RC0805FR-102K7L	Volume Production	DigiKey	13-RC0805FR-102K7LCT-ND	0.1	0.1
	10k	RES 10kohm ±1% 0.125W 2012 (0805 INCH)	R5, R7, R18, R19, R23, R25			6	0	Yageo Group	RC0805FR-0710KL	Volume Production	RS	6185163	0.00607	0.03643
	18k	RES 18kohm ±1% 0.125W 2012 (0805 INCH)	R22, R27			2	0	Yageo Group	RC0805FR-0718KL	Volume Production	Newark	49AK3089	0.005	0.01
	19k1	RES 19k1 OHM 5% 1/4W 0805	R30			1	0	Yageo Group	RC0805FR-0719K1L	Volume Production	Farnell	4061021	0.10977	0.10977
	47R	RES 47ohm ±1% 0.125W 2012 (0805 INCH)	R8, R9, R10, R11, R12, R13			6	0	Yageo Group	RC0805FR-0747RL	Volume Production	Newark	98K7855	0.004	0.024
	100k 1%	RES 100K OHM 1% 1/8W 0805 AC0805FR-13100KL	R17			1	0	Yageo Group	AC0805FR-13100KL	Volume Production	Farnell	4013022	0.02033	0.20332
	100k	RES 100kohm ±1% 0.125W 2012 (0805 INCH)	R29			1	0	Yageo Group	RC0805FR-07100KL	Volume Production	Mouser	603-RC0805FR-07100KL	0.086	0.086
	100R	RES 100ohm ±1% 0.125W 2012 (0805 INCH)	R6			1	0	Yageo Group	RC0805FR-07100RL	Volume Production	RS	6185135	0.00455	0.00455
	120k	RES 120kohm ±1% 0.125W 2012 (0805 INCH)	R20, R24			2	0	Yageo Group	RC0805FR-07120KL	Volume Production	Newark	49AK3045	0.004	0.008
	560R	RES 560ohm ±1% 0.125W 2012 (0805 INCH)	R1			1	0	Yageo Group	RC0805FR-07560RL	Volume Production	Newark	49AK3322	0.005	0.005
	680R	RES 680ohm ±1% 0.125W 2012 (0805 INCH)	R4			1	0	Yageo Group	RC0805FR-07680RL	Volume Production	Newark	49AK3363	0.005	0.005
	10k 1%	RES SMD 10K OHM 0.1% 1/8W 0805 AC0805FR-0710KL	R16			1	0	Yageo Group	AC0805FR-0710KL	Volume Production	Farnell	3495316	0.01784	0.17837
	10uH	Semi-Shielded Power Inductor, 10 uH, +/- 20%, 2.1 A, -40 to 125 degC, 2-Pin SMD, RoHS, Tape and Reel SRN3015-100M	L1			1	0	Bourns	SRN3015-100M	Volume Production	Farnell	2061634	0.41039	2.05
	1P1T-P	SW 600X600X700 24V 50mA THT	S1			1	0	TE Connectivity	1-1825910-4	Volume Production	Newark	95M4263	0.019	0.019
	5011	Test Point, Black, Through Hole, RoHS, Bulk	TP1, TP2, TP3, TP4			4	0	Keystone	5011	Volume Production	Newark	88H2535	0.137	0.548
	5010	Test Point, Red, Through Hole, RoHS, Bulk	TP5, TP6, TP7			3	0	Keystone	5010	Volume Production	Newark	62W5144	0.15	0.45
	SP0502BAHTG	TVS DIODE 5.5V 8.5V SOT23-3	D2			1	0	Littelfuse	SP0502BAHTG	Volume Production	Farnell	1827631	0.69853	0.69853
	SMBJ28A	TVS DIODE 28V 45.4V DO214AA	D3, D4, D6, D7, D9			5	0	Littelfuse	SMBJ28A	Volume Production	Farnell	1456531	0.25197	1.26
	LTST-C170KGKT	Ultra Bright AlInGaP Chip LED, 75 mW, -55 to 85 degC, 2-Pin SMD (0805), RoHS, Tape and Reel	D1			1	0	Lite-On	LTST-C170KGKT	Volume Production	DigiKey	160-1414-1-ND	0.18	0.18
	629712010214	USB 2.0 Type C Plug Vertical SMT	P1			1	0	Würth Elektronik	629712010214		DigiKey	732-629712010214CT-ND	4.55	4.55
	BZX84C3V6LT1G	Zener Voltage Regulator, 225 mW, 3-Pin SOT-23, Pb-Free, Tape and Reel	D5, D8			2	0	onsemi	BZX84C3V6LT1G	Volume Production	Farnell	2317468	0.11358	0.56789

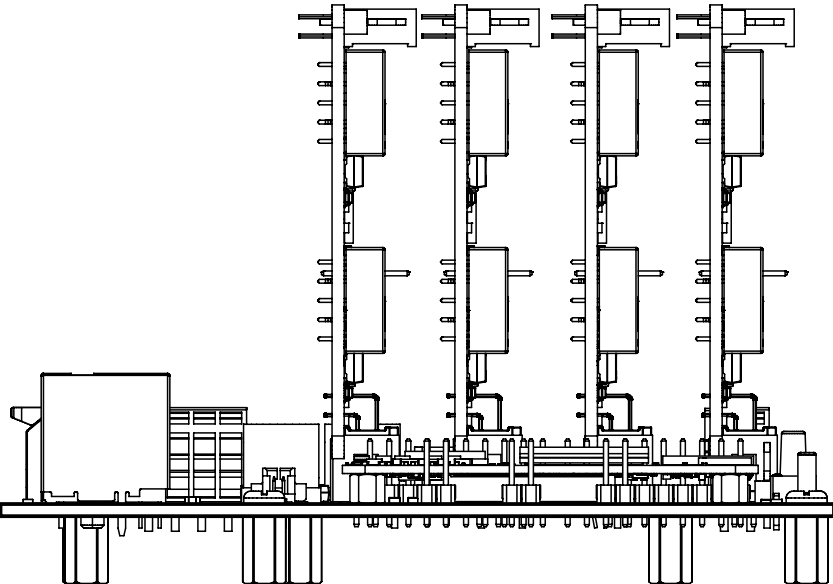
Top side (Scale 1:1)

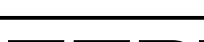


NOTES UNLESS OTHERWISE SPCEIFIED:

- 1. NOTE
- 2. NOTE
- 3. NOTE

Front side (Scale 1:1)



ETML Rue de Sébeillon 12 1004 Lausanne +41 316 77 77 www.etml.ch	Mod.	a	.	Dessiné	Léo Mendes	02.09.2025	A3		Echelle : 2:1
		b	.	Contrôlé	-	-			
		c	.				Nb. feuilles	Feuille n°	
		d	.	Remplace	.	=DRAF	4		1
	Nom du projet RegulationChauffage_Motherboard New						No 2510		

F

E

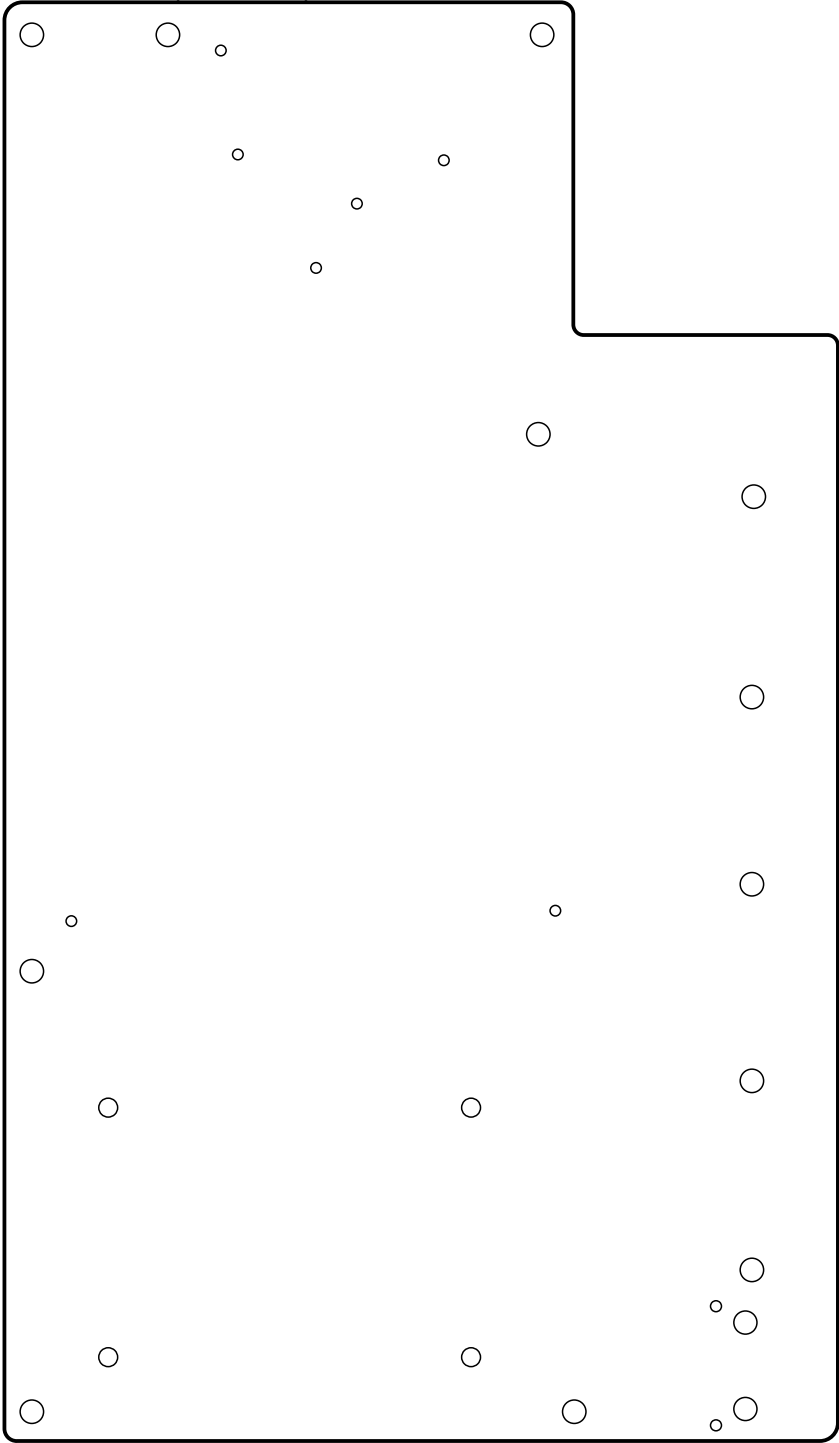
D

C

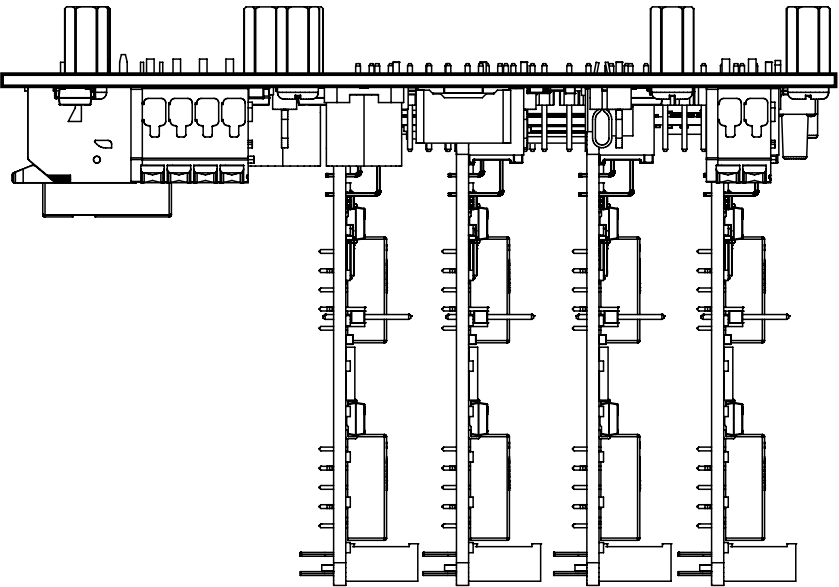
B

A

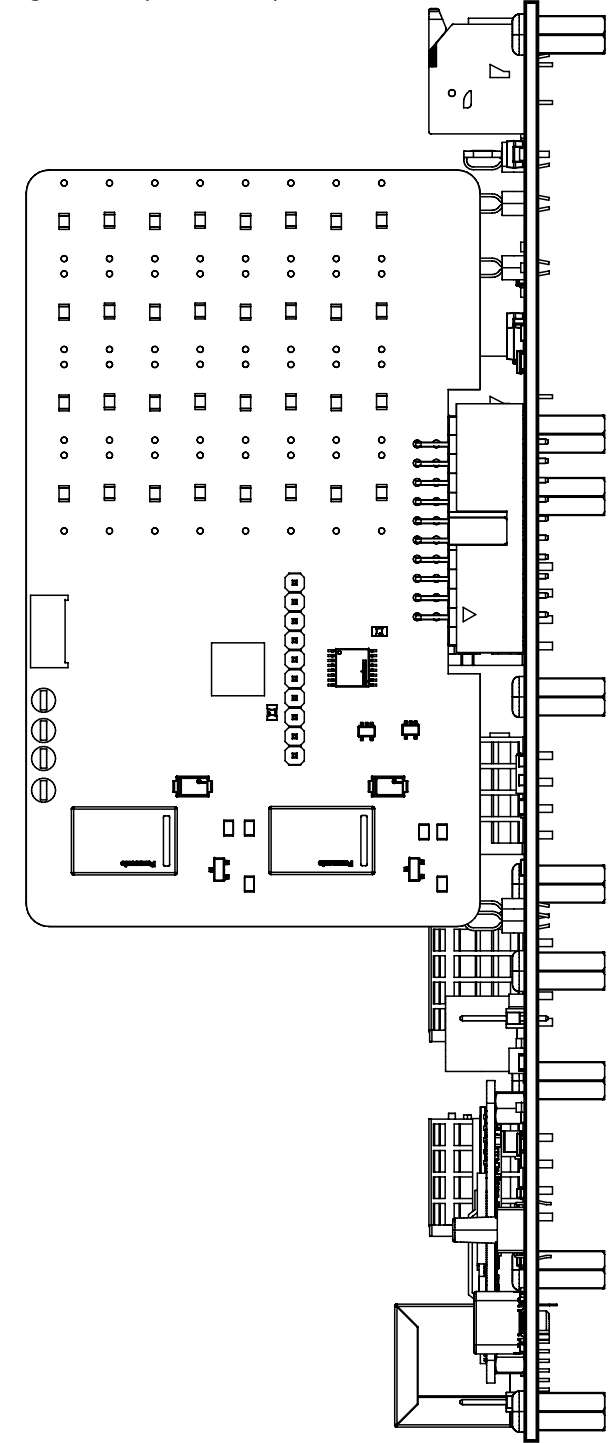
Bottom side (Scale 1:1)



Back side (Scale 1:1)



Right side (Scale 1:1)



ETML Rue de Sébeillon 12 1004 Lausanne +41 316 77 77 www.etml.ch	Mod.	a	.	Dessiné	Léo Mendes	02.09.2025	A3		Echelle : 2:1
		b	.	Contrôlé	-	-			
		c	.				Nb. feuilles	Feuille n°	
		d	.	Remplace	.	=DRAFT	4	2	
	Nom du projet RegulationChauffage_Motherboard						No 2510		

F

E

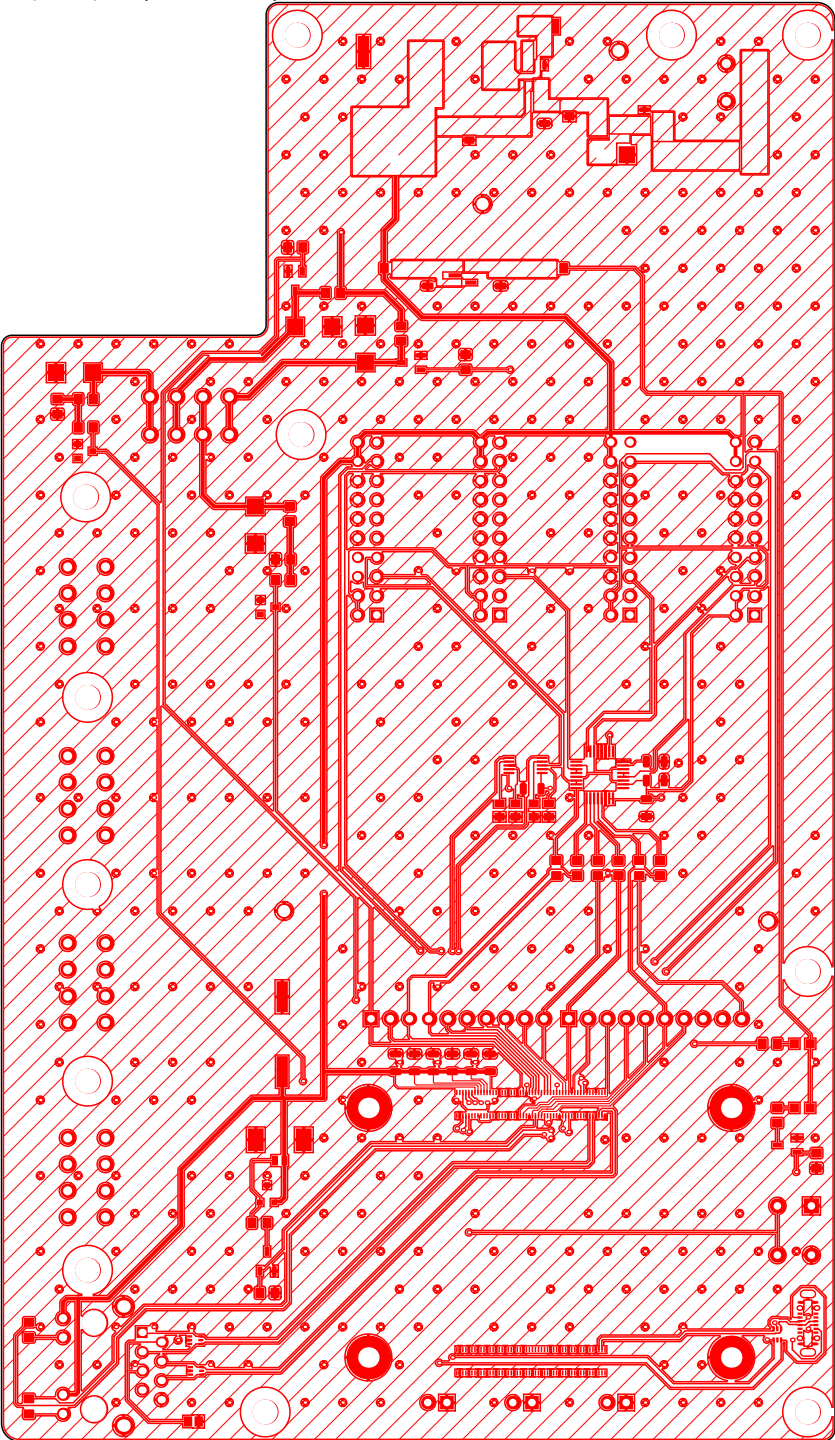
D

C

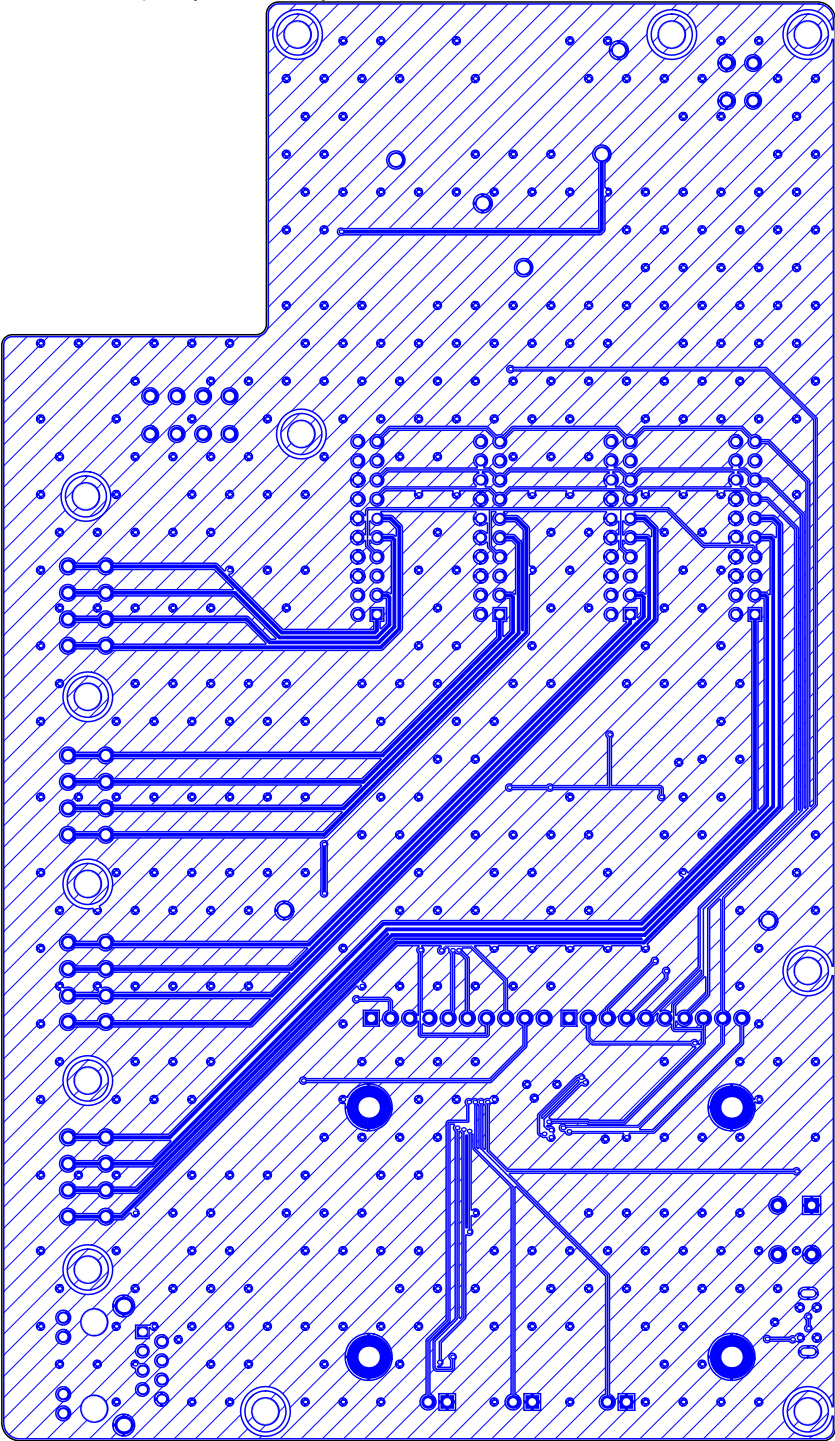
B

A

Top Layer (Scale 1:1)



Bottom Layer (Scale 1:1)



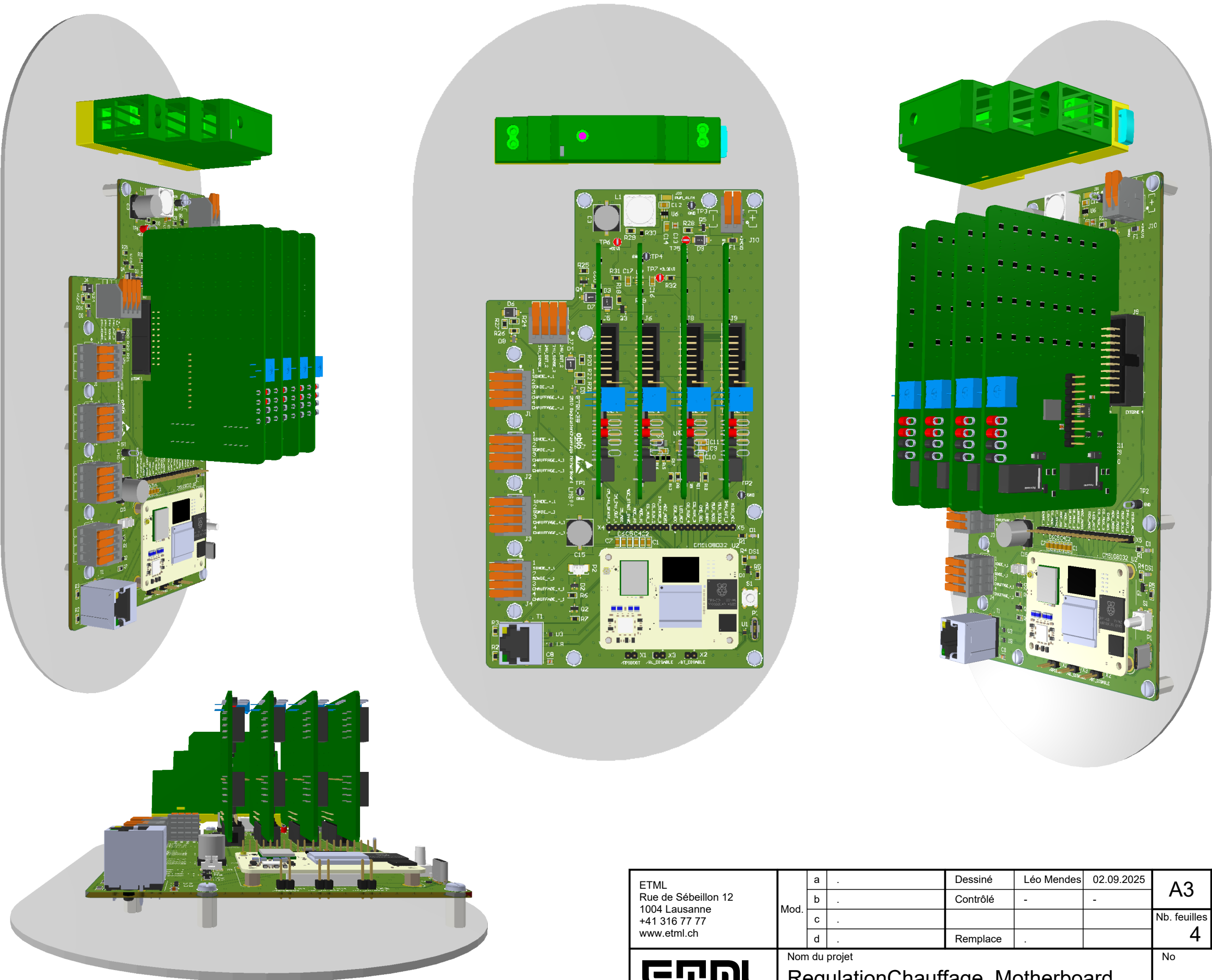
ETML Rue de Sébeillon 12 1004 Lausanne +41 316 77 77 www.etml.ch	Mod.	a	.	Dessiné	Léo Mendes	02.09.2025	A3		Echelle : 2:1
		b	.	Contrôlé	-	-			
		c	.				Nb. feuilles	Feuille n°	
		d	.	Remplace	.		4	3	
	Nom du projet RegulationChauffage_Motherboard						No 2510		

4

3

2

1



4

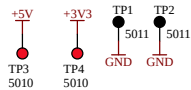
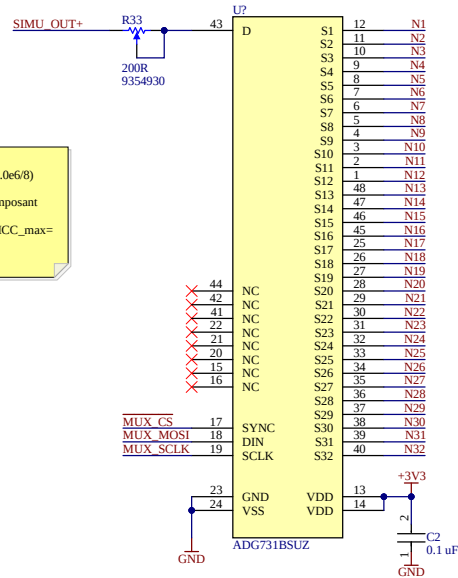
3

2

1

ETML Rue de Sébeillon 12 1004 Lausanne +41 316 77 77 www.etml.ch	Mod.	a	.	Dessiné	Léo Mendes	02.09.2025	A3		Echelle : 2:1
		b	.	Contrôlé	-	-			
		c	.				Nb. feuilles	Feuille n°	
		d	.	Remplace	.		4	4	
	Nom du projet						No		
	RegulationChauffage_Motherboard						2510		

F


$$\begin{aligned} I_{\text{dyn}} &= \text{CPD-VCC-fi} + \Sigma(\text{CL-VCC-fo}) \\ &= 115\text{e-}12 \cdot 3.3 \cdot 1.0\text{e}6 + 40\text{e-}12 \cdot 3.3 \cdot (1.0\text{e}6/8) \\ &= 0.3795\text{e-}3 + 0.0165\text{e-}3 \\ &= 0.3960\text{e-}3[\text{A}] = 0.396[\text{mA}] \text{ par composant} \\ I_{\text{total_max_par_composant}} &= I_{\text{dyn}} + I_{\text{CC_max}} = \\ &0.396[\text{mA}] + 0.080[\text{mA}] = 0.476[\text{mA}] \end{aligned}$$


Data: $V = 5.0$; $R_{coil} = 178$; $R_{68} = 0$; $R_{ds_on} = 0.08$
 $I = V / (R_{coil} + R_{ds_on}) = 5.0 / (178 + 0.08) = 0.0281 \text{ A}$
 $P_{total_5V} = V * I = 5.0 * 0.0281 = 0.1405 \text{ W}$
 $P_{coil} = I^2 * R_{coil} = 0.0281^2 * 178 = 0.1404 \text{ W}$
 $P_{MOSFET} = I^2 * R_{ds_on} = 0.0281^2 * 0.08 = 0.000063 \text{ W}$
 OFF state: $I \approx 0 \text{ A}$; $P \approx 0 \text{ W}$

[illegible]

ObloSolution * * * *	Mod.	a		Dessiné	LMS	19.08.2025	A3	 	Echelle :
		b		Contrôlé	*				
		c							
		d		Remplace					
							Nb. feuilles	Feuille n°	
							1	1	

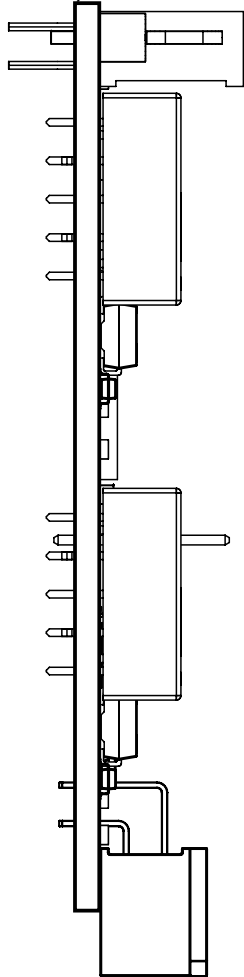
	Norm du projet	No
	RegulationChauffage	2510B

Line #	Name	Description	Designator	Revision ID	Revision Status	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
	0.1 uF	CAP CER 0.1 uF 10V X5R 0805	C2			1	Yageo Group	CC0805KRX7R9BB104	Volume Production	Farnell	3019949	0.03006	0.30062
	SP1002-02JTG	16V Pince 2 A (8/20µs) Ipp TVS - Diodes Montage en surface SC-70- 5	D1, D2			2	Littelfuse	SP1002-02JTG	Volume Production	DigiKey	F4238CT-ND	0.37	0.74
	SS14	DIODE SCHOTTKY 40V 1A SMA	D3, D4			2	onsemi	SS14	Volume Production	Farnell	1467537	0.26637	1.33
	SFH11-PBPC-D10-RA-BK	No Description Available	J1			1	Sullins	SFH11-PBPC-D10-RA-BK	Volume Production	Digikey	S9205-ND	1.29	1.29
	5019		J2, J3			2	Keystone	5019	Volume Production	DigiKey	36-5019CT-ND	0.37	0.74
	TQ2-5V-3	RELAY TELECOM DPDT 1A 5VDC	K1, K2			2	Panasonic	TQ2-5V-3	Volume Production	DigiKey	255-2257-ND	3.48	6.96
	PCB		PCB1			1	Eurocircuit	-	Manual Solution	Eurocircuit	-	77.08	77.08
	IRLML2402		Q1, Q2			2	UMWIRLML2402TR	UMWIRLML2402TR	Manual Solution	Digikey	4518-UMWIRLML2402TR-ND	0.48187	0.96373
	820R	820 Ohms ±1% 0,125W, 1/8W Pavé résistif 0805 (2012 métrique) Couche épaisse - RC0805FR- 13820RL	R1			1	Yageo Group	RC0805FR-13820RL	Volume Production	DigiKey	13-RC0805FR-13820RLCT-ND	0.1	0.1
	6R2	RES 6.2 OHM 1% 1/8W 0805 RC0805FR-076R2L	R2, R3, R5, R7, R10, R14			6	Yageo Group	RC0805FR-076R2L	Volume Production	Farnell	3952295	0.03667	0.36673
	6R8	RES 6.8 OHM 1% 1/8W 0805	R4, R6, R8, R9, R11, R12, R13, R15, R16, R17, R18, R19, R20, R21, R22, R24, R25, R26, R28, R29, R31			21	Yageo Group	RC0805FR-076R8L	Volume Production	DigiKey	311-6.80CRCT-ND	0.019	0.399
	7R5	RES SMD 7.5 OHM 1% 1/8W 0805 AC0805FR-077R5L	R23, R27, R30, R32			4	Yageo Group	AC0805FR-077R5L	Volume Production	DigiKey	13-AC0805FR-077R5LCT-ND	0.027	0.27
	200R	200 Ohms 0.5W, 1/2W PC Pins Through Hole Trimmer Potentiometer Cermet 1.0 Turn Side Adjustment	R33			1	Bourns	3386H-1-201LF	Volume Production	Farnell	9354930	1.78	1.78
	100R	RES 100ohm ±1% 0.125W 2012 (0805 INCH)	R34, R35			2	Yageo Group	RC0805FR-07100RL	Volume Production	Farnell	9237364	0.01661	0.16607
	0R	RES 0ohm ±1% 0.125W 2012 (0805 INCH)	R36, R38			2	Vishay	CRCW08050000Z0EA	Volume Production	DigiKey	541-0.0ACT-ND	0.02222	0.2222
	100k	RES 100kohm ±1% 0.125W 2012 (0805 INCH)	R37, R39			2	Yageo Group	RC0805FR-07100KL	Volume Production	Farnell	9237879	0.01555	0.15554
	5011	Test Point, Black, Through Hole, RoHS, Bulk	TP1, TP2			2	Keystone	5011	Volume Production	Farnell	2431148	0.14468	0.28936
	5010	Test Point, Red, Through Hole, RoHS, Bulk	TP3, TP4			2	Keystone	5010	Volume Production	DigiKey	36-5010-ND	0.39	0.78
	ADG731BSUZ	32-Channel, Serially Controlled 3.5Î© 1.8 V to 5.5 V, Â±2.5 V, Analog Multiplexer	U?			1							

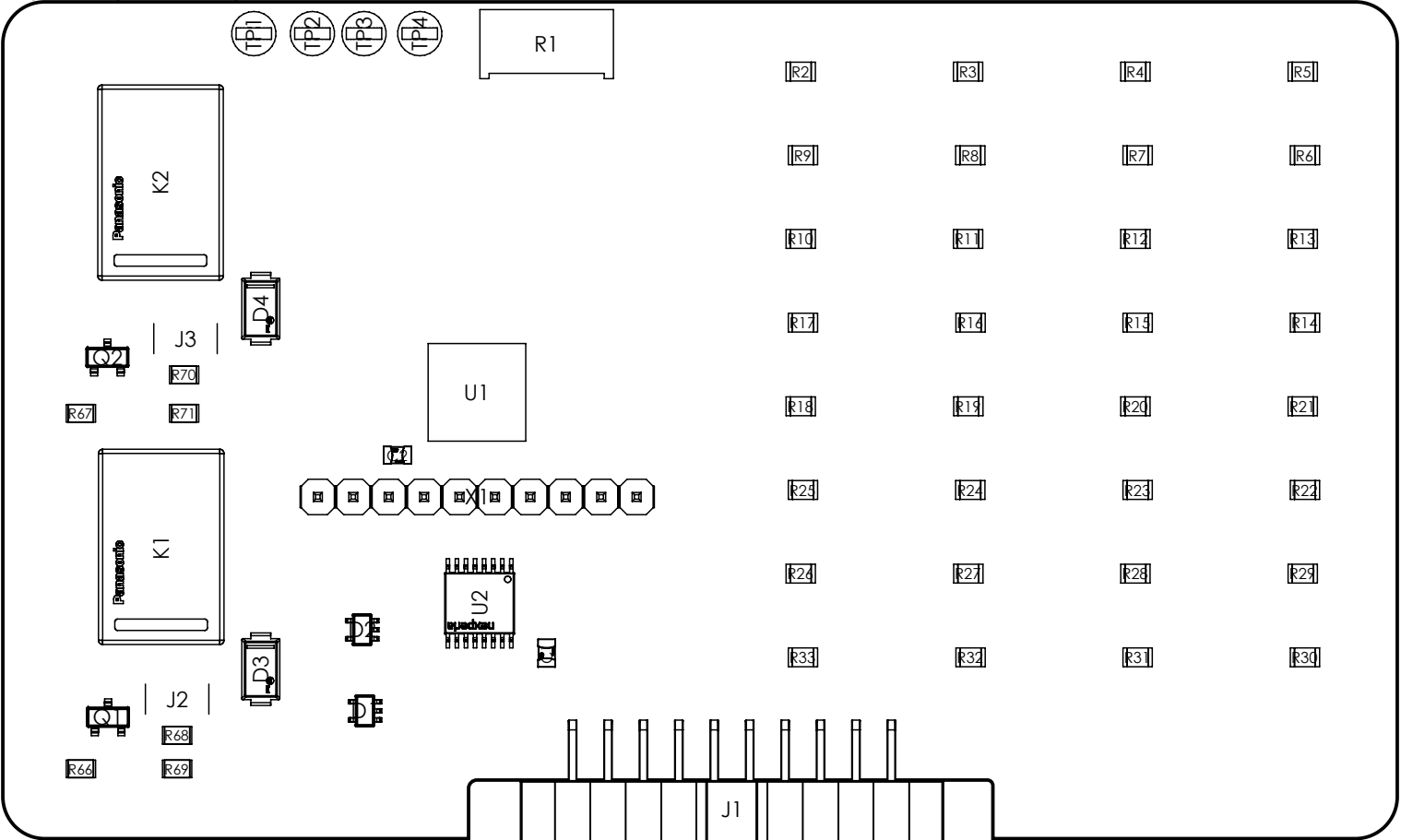
NOTES UNLESS OTHERWISE SPCEIFIED:

- 1. NOTE
- 2. NOTE
- 3. NOTE

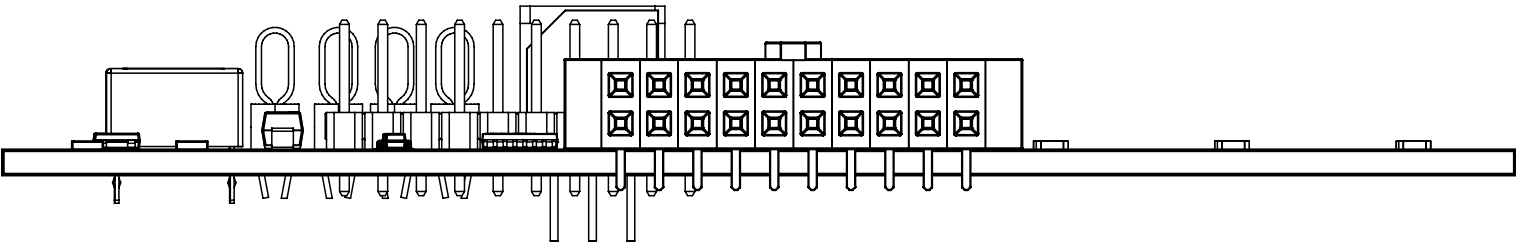
Left side (Scale 2:1)



Top side (Scale 2:1)



Front side (Scale 2:1)



ETML Rue de Sébeillon 12 1004 Lausanne +41 316 77 77 www.etml.ch	Mod.	a	.	Dessiné	Léo Mendes	02.09.2025	A3		Echelle : 2:1
		b	.	Contrôlé	-	-			
		c	.				Nb. feuilles	Feuille n°	
		d	.	Remplace	.	=DRAFT	4	1	
	Nom du projet RegulationChauffage_Daugtherboard New						No 2510		

F

E

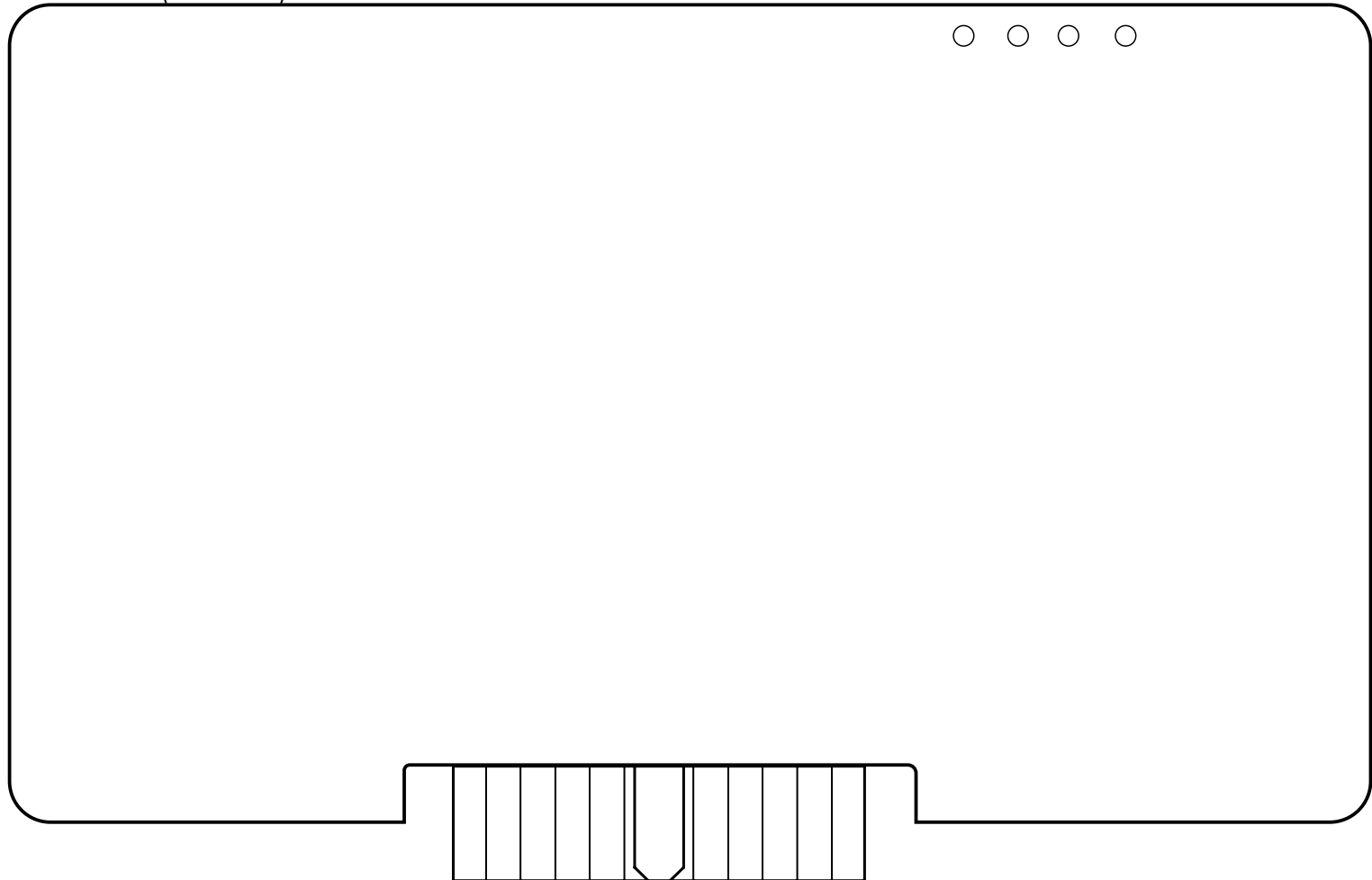
D

C

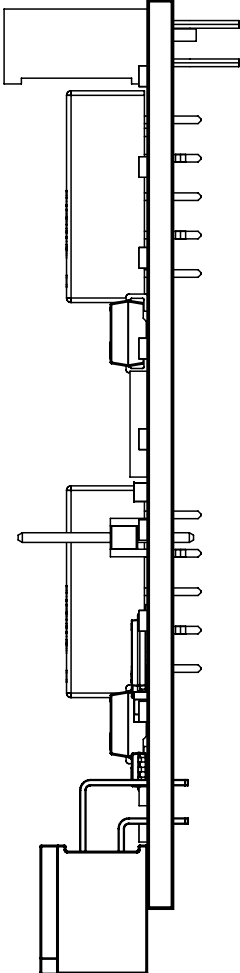
B

A

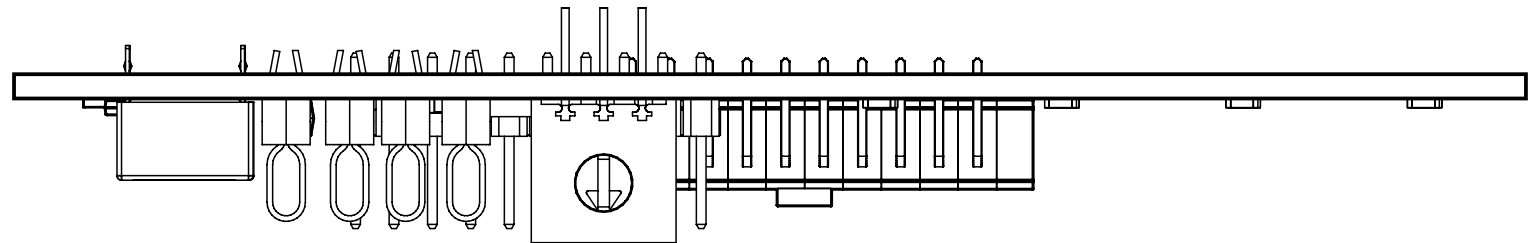
Bottom side (Scale 2:1)



Right side (Scale 2:1)



Back side (Scale 2:1)



ETML Rue de Sébeillon 12 1004 Lausanne +41 316 77 77 www.etml.ch	Mod.	a	.	Dessiné	Léo Mendes	02.09.2025	A3		Echelle : 2:1
		b	.	Contrôlé	-	-			
		c	.				Nb. feuilles	Feuille n°	
		d	.	Remplace	.	=DRAFT	4	2	
	Nom du projet						No		
	RegulationChauffage_Daugtherboard						2510		

F

E

D

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B

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E

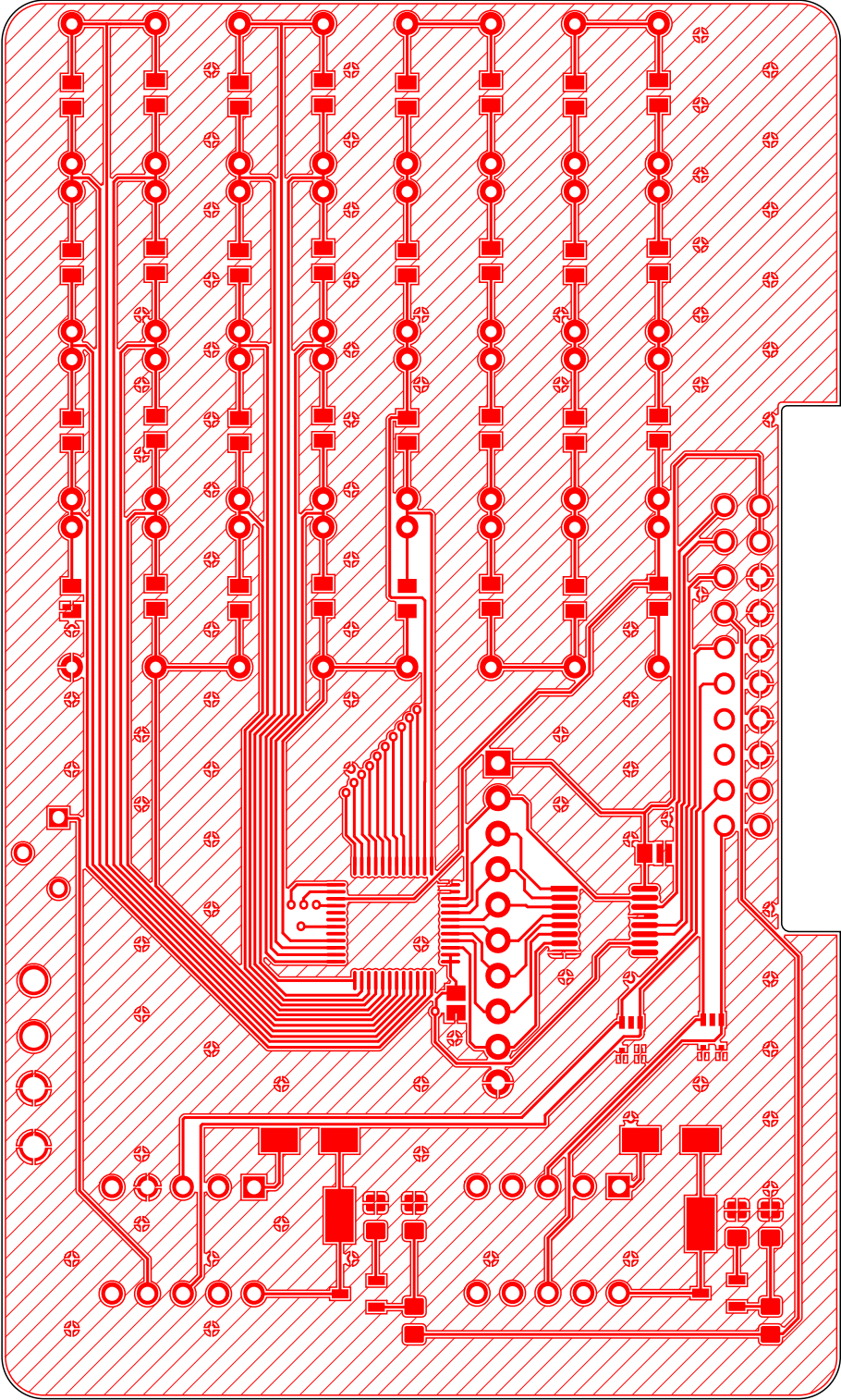
D

C

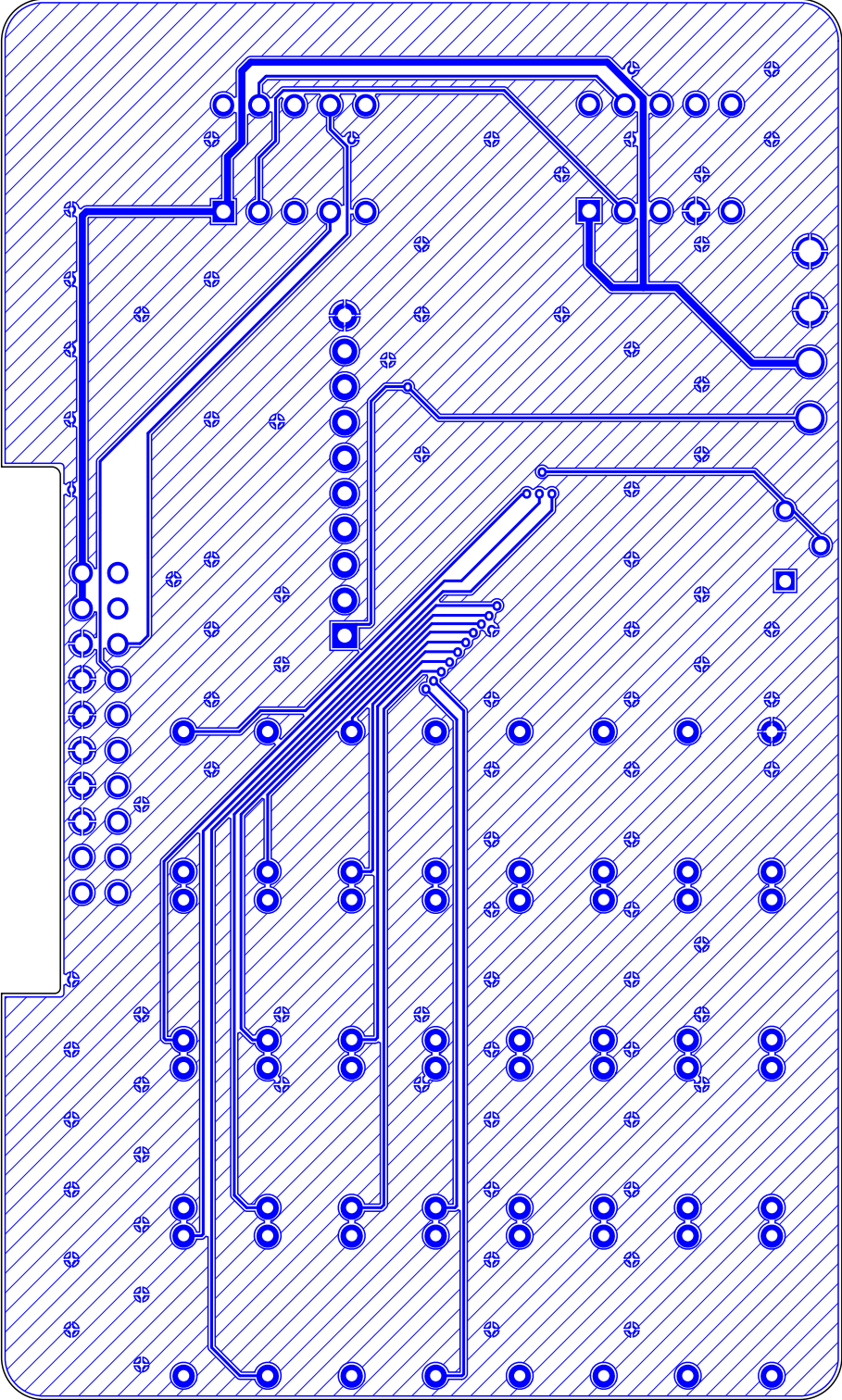
B

A

Top Layer (Scale 2:1)



Bottom Layer (Scale 2:1)



ETML Rue de Sébeillon 12 1004 Lausanne +41 316 77 77 www.etml.ch	Mod.	a	.	Dessiné	Léo Mendes	02.09.2025	A3		Echelle : 2:1
		b	.	Contrôlé	-	-			
		c	.				Nb. feuilles	Feuille n°	
		d	.	Remplace	.		4	3	
	Nom du projet RegulationChauffage_Daugtherboard						No 2510		

F

E

D

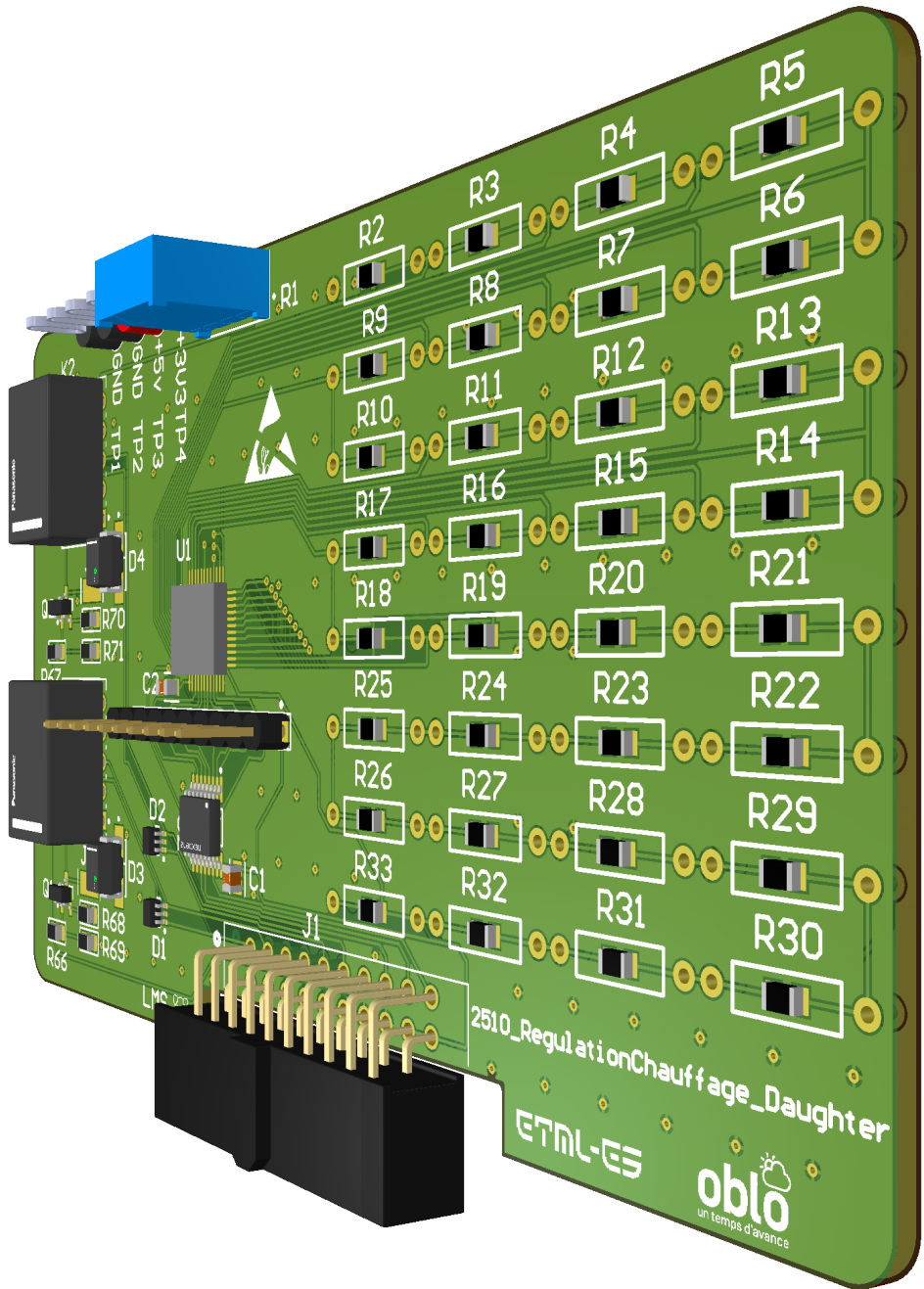
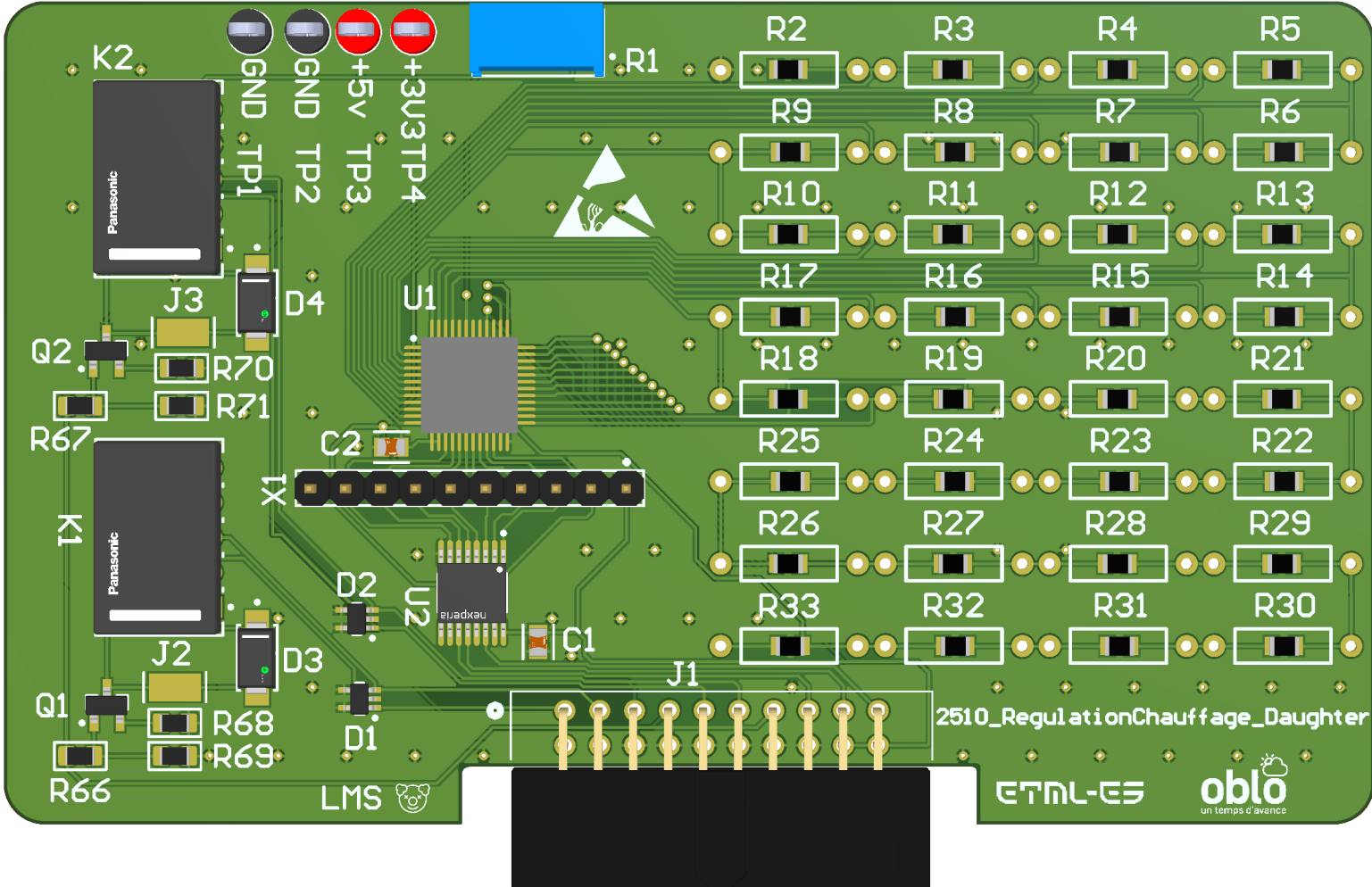
C

B

A

Realistic View

Realistic View



Mod.	a	.	Dessiné	Léo Mendes	02.09.2025	A3		Echelle : 2:1
	b	.	Contrôlé	-	-			
	c	.				Nb. feuilles	Feuille n°	
	d	.	Remplace	.		4	4	
ETML		Nom du projet RegulationChauffage_Daugtherboard				No 2510		